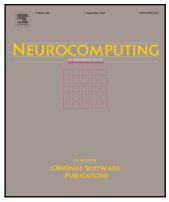
Neurocomputing 610 (2024) 128534



Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom





Shape-intensity-guided U-net for medical image segmentation

Wenhui Dong, Bo Du, Yongchao Xu *

National Engineering Research Center for Multimedia Software, Wuhan University, Wuhan, China
Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan, China
Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan, China

ARTICLE INFO

Communicated by G. Yang

Keywords:
Medical image segmentation
Texture bias
Shape-intensity prior
Model generalization

ABSTRACT

Medical image segmentation has achieved impressive results thanks to U-Net or its alternatives. Yet, most existing methods perform segmentation by classifying individual pixels, tending to ignore the shape-intensity prior information. This may yield implausible segmentation results. Besides, the segmentation performance often drops greatly on unseen datasets. One possible reason is that the model is biased towards texture information, which varies more than shape information across different datasets. In this paper, we introduce a novel Shape-Intensity-Guided U-Net (SIG-UNet) for improving the generalization ability of variants of U-Net in segmenting medical images. Specifically, we adopt the U-Net architecture to reconstruct class-wisely averaged images that only contain the shape-intensity information. We then add an extra similar decoder branch with the reconstruction decoder for segmentation, and apply skip fusion between them. Since the class-wisely averaged image has no any texture information, the reconstruction decoder focuses more on shape and intensity features than the encoder on the original image. Therefore, the final segmentation decoder has less texture bias. Extensive experiments on three segmentation tasks of medical images with different modalities demonstrate that the proposed SIG-UNet achieves promising intra-dataset results while significantly improving the cross-dataset segmentation performance. The source code will be publicly available after acceptance.

1. Introduction

Medical image segmentation [1–22,22] is a fundamental task in medical image analysis. The goal is to segment structures of interest, such as organs, tissues, or lesions. Accurate segmentation of medical images is crucial for diagnosis, treatment planning, and outcome evaluation in clinical usage.

Medical image segmentation has achieved remarkable performance thanks to deep learning [23–35]. Since the pioneer encoder–decoder U-Net [11] architecture, various modifications and improvements to this basic architecture have been proposed in much literature [15,17,36,37] to further enhance the performance of the segmentation models. For instance, U-Net++ [17] improves the U-Net architecture by using nested and dense skip pathways, facilitating semantic similarity between the feature maps of the encoder and decoder sub-networks. Though there are many variants of U-Net, Gut et al. [38] found that most of them do not exhibit consistent superiority over the classical U-Net via systematically evaluating multiple U-Net variants.

Some studies [39,40] have discovered that the convolutional neural networks (CNNs) have a tendency to focus more on learning detailed texture information for image analysis, Eliminating this tendency to learn textures can significantly improve the model's robustness and generalizability. Some studies [41,42] have shown that while Vision Transformers (ViTs) are less prone to learning texture information compared to CNNs, this issue still persists in ViTs. Training processes that focus more on shape features can enhance the performance and generalizability of the model. For medical images, the texture information of the target often varies considerably across different medical datasets. Yet, the object of interest in medical images usually has a specific shape and intensity. The texture bias model may lead to overfitting on the intra-dataset, resulting in poor generalization performance on the cross-dataset. Considering that the shape of organs or lesions usually has certain inherent anatomical structures, many methods (see [43] for a comprehensive survey) have been proposed to enhance the shape prior knowledge in the segmentation network. For instance, some methods [44–46] incorporate shape information by adding edge constraints to improve the segmentation accuracy. Some methods [47,48] employ an autoencoder network to refine the predicted segmentation or fuse the latency space feature of the autoencoder network to the segmentation network to incorporate shape information into the segmentation network.

Although they have shown promising results in enhancing segmentation outcomes, they are not entirely effective in addressing the issue

* Corresponding author at: Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan, China
E-mail address: yongchao.xu@whu.edu.cn (Y. Xu).

https://doi.org/10.1016/j.neucom.2024.128534
Received 21 March 2024; Received in revised form 28 August 2024; Accepted 1 September 2024
Available online 3 September 2024
0925-2312/© 2024 Published by Elsevier B.V.

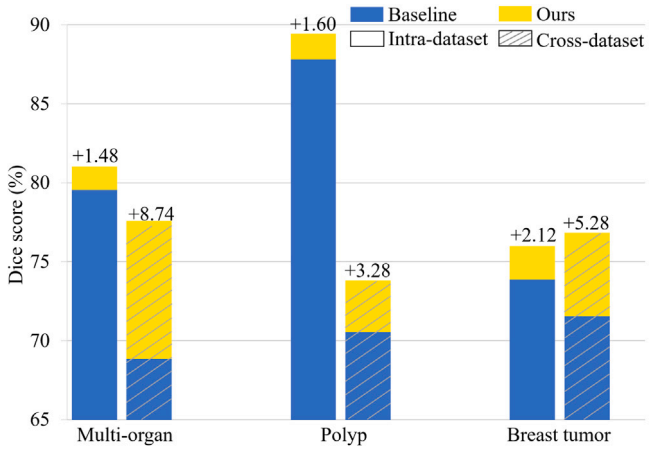


Fig. 1. Comparison of averaged performance between the proposed method and some corresponding baseline models (including U-Net [11], the baseline of SAUNet [44], nnU-Net [37], SANet [49], LM-Net [50] and 2D D-LKA Net [51]) under intra-dataset and cross-dataset evaluation on three medical image segmentation tasks.

of network bias towards learning textures. In this paper, we present a novel approach to directly alleviate the texture bias issue. More precisely, we reconstruct the class-wisely averaged image using the original U-Net, and add an extra decoder branch for segmentation. Since the class-wisely averaged image has no texture information, the reconstruction branch focuses on extracting the shape and intensity feature. We then leverage a shape-intensity attention block (SIAB) consisting of spatial attention and channel attention to better fuse shape-intensity features from the reconstruction branch into the segmentation branch. This results in accurate segmentation and good generalization across different images (see Fig. 1). The proposed approach is designed to address the challenge of texture bias in U-Net [11] and its variants, which tends to negatively impact segmentation performance on unseen datasets. The main contribution of this work is as follows:

(1) We novelly propose to reconstruct class-wisely averaged images without texture information for extracting shape-intensity features; (2) We leverage a shape-intensity attention block to incorporate shape and intensity features of the reconstruction decoder into the segmentation decoder, effectively alleviating the issue of texture bias; (3) Without bells and whistles, SIG-UNet achieves noticeable improvements over the corresponding baseline methods on three medical image segmentation tasks with different imaging modalities. Furthermore, the proposed method has a much better generalization ability to unseen datasets.

The rest of this paper is organized as follows. In Section 2, we review some related works on medical image segmentation and some incorporating shape information methods. We then detail the proposed method in Section 3, followed by extensive experimental results in Section 4 and some discussions in Section 5. Finally, we conclude and give some perspectives in Section 6.

2. Related work

We shortly review some alternatives of U-Net in Section 2.1, and comprehensively review some methods that incorporate shape prior into medical image segmentation in Section 2.2.

2.1. Medical image segmentation

With the advancement of deep learning, the task of medical image segmentation has witnessed significant development over the past decades. The U-Net family [11–17] is known as a powerful architecture for addressing medical image processing challenges. The skip-connections in U-Net or its alternative networks effectively combine shallow features with deep features so that they perform well most of the time. Xiang et al. [52] and Alom et al. [53,54] propose recurrent convolutional neural networks that incorporate backward skip-connections from the decoder to the encoder, effectively passing semantics between the two parts. These architectures achieve satisfactory performance without the need for additional parameters. Attention U-Net [36] introduces additional gated attention mechanism branches to fuse encoded and decoded features at the same level. Res-UNet [55] enhances the performance of retinal vessel segmentation through the incorporation of a weighted attention mechanism. 3D-Unet [15] and V-Net [14] are well-known methods for 3D medical image segmentation. LM-Net [50] introduces a multi-scale lightweight U-shape architecture that integrates the advantages of convolutional neural networks and vision transformers. D-LKA Net [51] incorporates a deformable large kernel attention mechanism that significantly enhances the model's feature extraction capabilities.

Although these alternatives of U-Net have improved the segmentation performance of original U-Net to some extent, they often overlook the network's bias towards learning texture information [39,40], limiting their ability to generalize well on unseen data. To address this issue, we propose to leverage U-Net to reconstruct class-wisely averaged images, allowing the network to focus more on shape information. Subsequently, we integrate the shape features of the reconstruction branch into the newly added segmentation branch to enhance the segmentation and generalization ability.

2.2. Incorporation of shape prior into segmentation

Integrating prior knowledge into the segmentation network has proven useful to obtain more accurate segmentation results [43–48,56–67]. In addition to the classical pixel-wise segmentation loss (*e.g.*, cross-entropy loss and/or Dice loss), some methods [44–46,56] also adopt other losses to learn extra shape-related targets (usually based on the object boundary). This helps to learn shape-related feature and fuse it with segmentation features to implicitly incorporate shape information for segmentation. In particular, SAUNet [44] introduces a shape extraction stream, which is composed of gated convolutional layers that perform edge prediction. SMU-Net [46] develops a shape-aware loss that assesses the proximity of the segmented region to the nearest object boundary and a position-aware loss that measures the distance from the center of the object.

Some methods [47,48,58,60] employ an autoencoder network to refine the predicted segmentation or fuse the latency space feature of the autoencoder network to the segmentation network to incorporate shape information into the segmentation network. Chen et al. [60] also introduce a technique of hard example generation in the latent space of the segmentation network, which generates diverse training images and corrupted segmentation results to reinforce the performance of refined segmentation by the subsequent auto-reconstruction network. Painchaud et al. [58] propose a framework for cardiac image segmentation that utilizes a constrained variational autoencoder to learn a representation of valid cardiac shapes, allowing for the projection of anatomically implausible results into the cardiac latent space and steering them towards the closest valid shape. SGU-Net [48] appends an autoencoder after the segmentation network to impose shape constraints on both the predicted results and the ground truth.

There are also some methods [61,62,68] that adopt the method of adversarial learning to incorporate shape information into the segmentation network. Specifically, LMISA [62] learns a consistent potential feature representation with shape awareness from both domains by imposing a shape constraint using adversarial learning. Zhang et al. [68] use a shape consistency loss acting on two segmentation networks to provide implicit shape constraints on the anatomy.

The existing methods either introduce shape-related constraints or employ adversarial learning to improve the segmentation performance

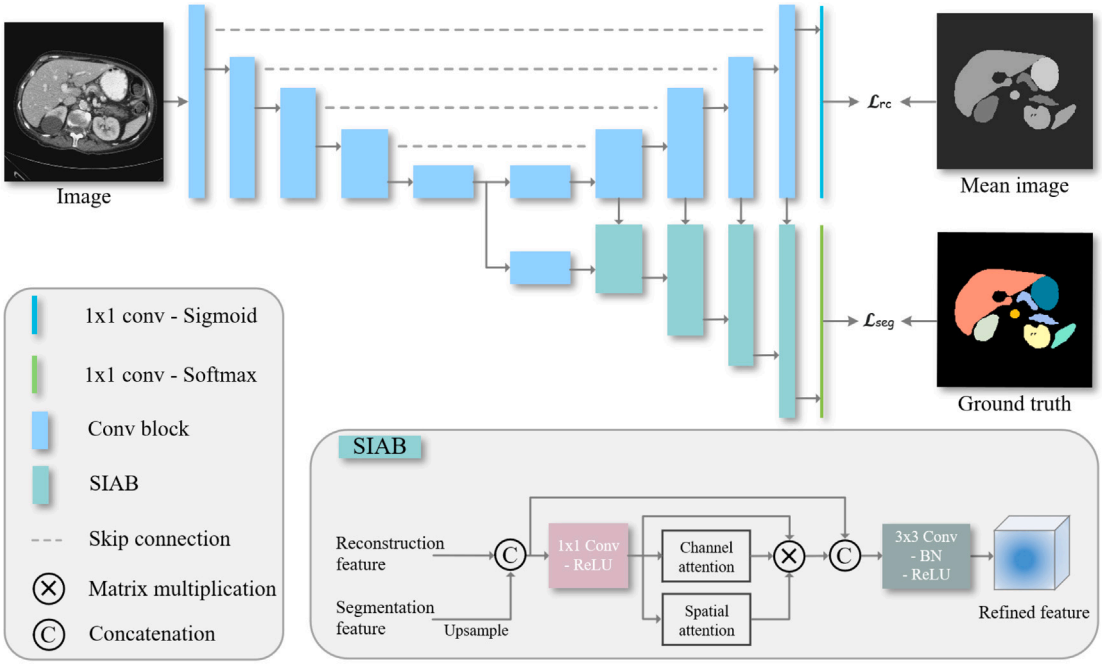


Fig. 2. The pipeline of the proposed shape-intensity-guided U-Net (SIG-UNet). The shape-intensity attention block (SIAB) is developed to fuse the texture inattentive feature into the segmentation branch. The Conv block is a sequence of Conv-batch norm (BN)-ReLU.

of the segmentation networks. In contrast to these methods, we propose a novel approach that leverages U-Net to reconstruct class-wisely averaged images for capturing shape and intensity information and alleviating the texture bias issue. We further introduce a segmentation branch for segmentation, along with an attention-based feature fusion module, which combines the features from the reconstruction branch into the segmentation branch. This effectively improves the segmentation accuracy and generalizability.

3. Method

3.1. Overview

Neural networks, especially convolutional neural networks, to prioritize texture information over shape in object recognition. Researches [39–41] indicate that this bias towards texture can adversely affect the robustness and generalizability of models. Furthermore, this texture bias ignores the fact that objects of interest in medical images usually have a specific shape distribution. We aim to alleviate such texture bias issue and incorporate shape and intensity prior into the segmentation. The proposed method is built upon variants of U-Net with an encoder–decoder and a skip connection architecture. The overall pipeline is depicted in Fig. 2. Specifically, the proposed method is composed of an encoder and two decoders: the reconstruction decoder and the segmentation decoder. The reconstruction decoder is used to extract shape and intensity features by reconstructing the class-wisely averaged image (see Section 3.2). Then a shape-intensity attention block (SIAB) is developed to incorporate the shape and intensity feature into the segmentation decoder (see Section 3.3).

3.2. Shape and intensity feature extraction

Medical objects often have specific shapes and intensities with varied textures across different images. Alleviating the bias to texture information and incorporating shape and intensity information into segmentation can improve the accuracy and generalization ability of the segmentation network. To extract the shape and intensity information, we propose to reconstruct the class-wisely averaged images (*i.e.* mean image in Fig. 2), which mainly have shape and intensity information without any texture information. Therefore, the reconstruction decoder serves as the shape and intensity feature extraction.

More precisely, for a given image f, we calculate the class-wisely averaged image f_m by taking the mean value of the pixel values within each class. Formally, let X_k denotes the set of pixels belonging to the kth class. For each pixel $x \in X_k$, the class-wisely averaged value $f_m(x)$ is given by:

$$f_m(x) = \sum_{x \in X_k} f(x)/|X_k|, \tag{1}$$

where $|\cdot|$ denotes the cardinality. The process is summarized in Algorithm 1. We apply Min-Max normalization to the input image, and adopt the Sigmoid operation to reconstruct the class-wisely averaged image. The training objective for the reconstruction branch $\mathcal{L}_{rc}$ is given by:

$$\mathcal{L}_{rc} = MSE(f_m, \hat{f}_m), \tag{2}$$

where MSE$(\cdot)$ is the Mean-Squared Error loss function, $\hat{f}_m$ is the output of the reconstruction branch.

3.3. Shape-intensity attention block

In order to effectively incorporate the shape and intensity feature of the reconstruction decoder into segmentation, we leverage a shape-intensity attention block (SIAB), encompassing a spatial and a channel attention module. Concretely, as shown in Fig. 2, the segmentation branch's feature is upsampled and concatenated with the corresponding feature from the reconstruction branch. Then, a 1×1 Conv-ReLU block is utilized to fuse these features, resulting in the intermediate feature map F, which serves as the input for both the channel and spatial attention modules.

Channel attention map: As shown in Fig. 3, we first aggregate spatial information of the feature map F by using a global average-pooling (denoted as $AvgPool$). The aggregated feature is initially fed into a 1×1 Conv-ReLU block to perform feature aggregation along the channel axis and followed by a 1×1 convolution to restore the number of channels (denoted as Φ_c). Finally, the features pass through

Algorithm 1: Transform images by averaging pixel values within class regions

Input: Dataset Ω containing N images and corresponding labels, number of classes K

Return: Set Φ containing images where each class region is filled with its mean pixel value

```
for n = 1 : N do
    f_n, l_n ∈ Ω   // the n-th image and the corresponding label
    for k = 1 : K do
        mask_k = (l_n == k)   // obtain the mask of class k
        intensity_k = mean(f_n[mask_k])   // calculate mean pixel value
        f_n[mask_k] = intensity_k   // fill the region with its mean value
    Φ.append(f_n)   // add the modified image to Φ
Return: Φ
```

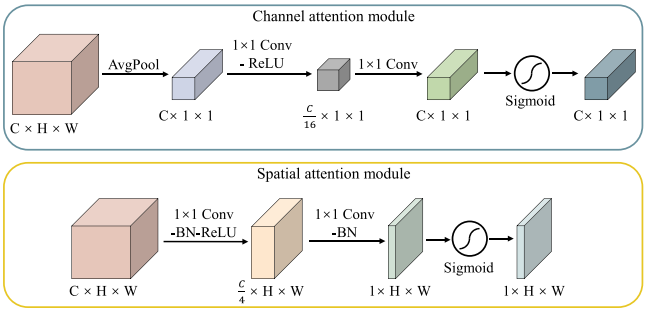


Fig. 3. Illustrations of channel attention module and spatial attention module. As depicted in the figure, C, H, W respectively represent the number of channels, height, and width. The channel attention module utilizes average pooling outputs and two consecutive convolution operations to derive channel attention features. The spatial attention module employs two successive convolutions along the channel axis to compress and produce the spatial attention map.

the sigmoid function to obtain the channel attention map $M_c \in \mathbb{R}^{C\times1\times1}$. In short, the channel attention is computed as:

$$M_c(F) = \sigma(\Phi_c(AvgPool(F))), \tag{3}$$

where σ denotes the sigmoid function.

Spatial attention map: As shown in Fig. 3, the feature map F is initially processed through a sequence of 1 × 1 Conv-BN-ReLU, which compresses the spatial dimensions along the channel axis to one-quarter of their original size. Subsequently, a 1 × 1 Conv-BN block further reduces the dimensions to one. Finally, a sigmoid function is applied to generate the spatial attention map $M_s \in \mathbb{R}^{1\times H\times W}$. Formally, it is given by:

$$M_s(F) = \sigma(\Phi_s(F)), \tag{4}$$

where Φ_s represents the parameters corresponding to the two compression stages.

We simply fuse the feature F, the attention map M_c and M_s by using element-wise multiplication. Note that we broadcast the channel (*resp.* spatial) attention values along the spatial (*resp.* channel) dimension during the multiplication. The fused feature is then concatenated with the reconstruction feature, followed by a Conv block to generate the final refined feature.

We adopt the classical combination of cross-entropy loss and Dice loss as the training objective $\mathcal{L}_{seg}$ for the segmentation branch. The overall loss $\mathcal{L}$ for the whole pipeline is given by:

$$\mathcal{L} = \mathcal{L}_{seg} + \alpha \times \mathcal{L}_{rc}, \tag{5}$$

where α is a hyper-parameter to balance the contribution of the two loss terms and is set to 1 for all experiments.

4. Experiments

4.1. Datasets and evaluation protocols

4.1.1. Multi-organ segmentation in CT images

We conduct experiments on the synapse multi-organ segmentation dataset[1] [69] involved in the MICCAI 2015 Multi-Atlas Abdomen Labeling Challenge. This dataset contains 3779 axial contrast-enhanced images from 30 abdominal CT scans, where 18 (*resp.* 12) cases are used for training (*resp.* testing). Each image slice is of size 512×512. The goal is to segment 8 abdominal organs: aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen, and stomach. We conduct evaluations on the AbdomenCT-1K dataset[2] [70] and FLARE 2021 [71] to benchmark the generalization ability of different methods. AbdomenCT-1K is a large-scale dataset containing more 1000 CT scans from 12 medical centers for liver, kidney, spleen, and pancreas segmentation. FLARE 2021 is also a CT dataset that includes these four abdominal organs and is collected from 11 medical centers.

4.1.2. Polyp segmentation in colonoscopy images

Following the experimental setup in [72], we conduct experiments on five public datasets for colorectal polyp segmentation, including Kvasir [73], CVC-ClinicDB [74], CVC-ColonDB [75], ETIS [76], and Endoscene [77]. We randomly split Kvasir and CVC-ClinicDB dataset into 4:1 for training and testing, respectively. We also perform evaluations on the other three datasets to assess the generalization ability of different methods. Note that the Endoscene dataset is a combination of CVC-ClinicDB and CVC-300. We only evaluate the model on the test set of CVC-300 denoted as CVC-T. For all methods, we report results of the best model of different training epochs on the test set of Kvasir and CVC-ClinicDB.

4.1.3. Breast tumor segmentation in ultrasound images

We conduct breast tumor segmentation experiments on the BUSI dataset [78]. This dataset is collected from breast ultrasound images of 600 female patients between the age of 25 and 75 years old. It consists of a total of 780 images with an average image size of 500 × 500 pixels, including 487 benign images, 210 malignant images, and 133 normal images. This dataset is randomly split into 4:1 for training and testing. We conduct the cross-dataset evaluation on the dataset used in [79] and the dataset B [80] to benchmark the generalization ability of different methods. The dataset in [79] is composed of 42 breast ultrasound (BUS) images provided by the Imaging Department of the First Affiliated Hospital of Shantou University, Shantou (China). The dataset B [80] consists of 163 images corresponding to 110 benign and 53 malignant lesion images, collected in 2012 from the UDIAT Diagnostic Centre of the Parc Taulí Corporation, Sabadell (Spain).

4.1.4. Evaluation protocols

We adopt two metrics: Dice score (Dice), Intersection over Union (IoU) which are widely used metrics in medical image segmentation to evaluate the proposed method. Let TP, FP, TN, FN denote the number of true positive, false positive, true negative, and false negative pixels of the segmentation result, respectively. The Dice score is defined as:

$$Dice = \frac{2\times TP}{FP + 2\times TP + FN}. \tag{6}$$

The IoU is given by:

$$IoU = \frac{TP}{FP + TP + FN}. \tag{7}$$

[1] https://www.synapse.org/#!Synapse:syn3193805/wiki/217789
[2] https://flare22.grand-challenge.org/

Table 1
Comparison on the Synapse multi-organ segmentation dataset and the AbdomenCT-1K dataset [70]. ↑ (*resp.* ↓) indicates the higher (*resp.* lower) the better. SAUNet' represents the reproduced results based on the baseline of the SAUNet [44].

Method	Intra-dataset		Cross-dataset			
	Synapse		AbdomenCT-1K		FLARE 2021	
	IoU↑	DICE↑	IoU↑	DICE↑	IoU↑	DICE↑
Att-UNet [36]	0.674 ± 0.011	0.780 ± 0.011	0.512 ± 0.026	0.590 ± 0.029	0.534 ± 0.028	0.611 ± 0.030
ResUnet [81]	0.668 ± 0.011	0.778 ± 0.013	0.545 ± 0.022	0.632 ± 0.025	0.550 ± 0.024	0.641 ± 0.026
TransUNet [12]	0.693 ± 0.007	0.775 ± 0.008	0.606 ± 0.020	0.695 ± 0.021	0.617 ± 0.022	0.719 ± 0.023
U-Net [11]	0.665 ± 0.013	0.759 ± 0.013	0.456 ± 0.036	0.504 ± 0.037	0.463 ± 0.038	0.515 ± 0.039
SIG-UNet (Ours)	0.699 ± 0.008	0.790 ± 0.009	0.674 ± 0.026	0.745 ± 0.025	0.658 ± 0.028	0.740 ± 0.027
SAUNet' [44]	0.693 ± 0.013	0.782 ± 0.015	0.472 ± 0.026	0.556 ± 0.029	0.555 ± 0.028	0.658 ± 0.030
SAUNet [44]	0.694 ± 0.010	0.780 ± 0.012	0.517 ± 0.022	0.598 ± 0.025	0.570 ± 0.025	0.669 ± 0.027
SIG-SAUNet' (Ours)	0.715 ± 0.009	0.797 ± 0.008	0.678 ± 0.018	0.753 ± 0.019	0.690 ± 0.019	0.767 ± 0.022
nnUNet [37]	0.732 ± 0.009	0.810 ± 0.009	0.693 ± 0.015	0.786 ± 0.013	0.694 ± 0.015	0.788 ± 0.017
SIG-nnUNet (Ours)	0.739 ± 0.007	0.819 ± 0.006	0.728 ± 0.014	0.798 ± 0.017	0.736 ± 0.015	0.809 ± 0.016
LM-Net [50]	0.708 ± 0.012	0.793 ± 0.013	0.643 ± 0.015	0.727 ± 0.016	0.662 ± 0.015	0.751 ± 0.017
SIG-LMNet (Ours)	0.717 ± 0.011	0.806 ± 0.010	0.657 ± 0.016	0.759 ± 0.018	0.675 ± 0.013	0.765 ± 0.015
2D D-LKA Net [51]	0.766 ± 0.010	0.833 ± 0.009	0.719 ± 0.015	0.797 ± 0.016	0.728 ± 0.015	0.803 ± 0.018
2D SIG-DLKA (Ours)	**0.773 ± 0.008**	**0.839 ± 0.008**	**0.738 ± 0.013**	**0.809 ± 0.015**	**0.742 ± 0.015**	**0.814 ± 0.017**

4.2. Implementation details

Since the primary objective of this paper is not to achieve state-of-the-art performance on each segmentation task individually, we focus on demonstrating the effectiveness of the proposed SIG-UNet in integrating shape and intensity information for medical image segmentation. For that, we conduct experiments on several open-sourced segmentation networks on each corresponding segmentation task, including the most widely used U-Net [11] and nnU-Net [37] for medical image segmentation, SANet [49], LM-Net [50] and 2D D-LKA Net [51]. We also implement the proposed method on the baseline of SAUNet [44] (denoted as SAUNet' in the following), one of the very few shape-aware medical image segmentation methods that release the source code.

We benchmark the proposed method using both intra-dataset and cross-dataset evaluation settings on three medical image segmentation tasks with different imaging modalities. All experiments are conducted using the Pytorch framework on a workstation with a NVIDIA GeForce RTX 3090 GPU (24 GB). We employ the Adam optimizer [82] for training with a batch size of 16, conducting 200 training epochs. The model from the final epoch is selected for testing. We use the random seed and train five models to calculate the mean and standard deviation of performance metrics. In all experiments, our proposed method maintains the same training settings as the baseline, including classical data augmentation techniques such as random rotations and geometric affine transformations, to avoid overfitting, as well as the learning rate and its scheduling strategy. We use the unbiased estimation in our experimental analyses. Concretely, we set a random seed, and each experiment was repeated five times. The mean and standard deviation of the results are calculated and presented in the final experimental results table. In all experiments, the hyper-parameter α involved in Eq. (5) is set to 1.

4.3. Results

4.3.1. Multi-organ segmentation

We assess the effectiveness of the proposed method on Multi-organ segmentation in CT images. For multi-organ segmentation in CT images, following [12,13], we take each slice of the 3D volume as an input to the network and stack the predicted 2D slices together to reconstruct the 3D prediction for evaluation. The quantitative comparison is given in Table 1. SIG-UNet built on the U-Net improves the baseline model by 3.4% IoU and 3.1% Dice score The proposed method built on the baseline of SAUNet achieves similar improvement as SIG-UNet based on U-Net baseline. Our method outperforms the SAUNet by 2.1% IoU and 1.70% Dice score We also implement the proposed method based on the nnU-Net [37] denoted as SGnnU-Net. As shown in Table 1, SGnnU-Net outperforms the nnU-Net by 0.7% IoU and 0.9% Dice score. Employing the latest method LM-Net [50] and 2D D-LKA Net [51] as benchmarks, our approach continues to enhance the performance of the corresponding method.

In order to benchmark the generalization performance of the proposed approach, we evaluate the model on AbdomenCT-1K dataset [70] and FLARE 2021 [71]. As depicted in Table 1, the proposed method outperforms all the corresponding baseline models under cross-dataset evaluation. In particular, the proposed method based on U-Net demonstrates significant improvement with 21.8% (*resp.* 19.5%) IoU and 24.1% (*resp.* 22.5%) increase in Dice score on AbdomenCT-1K dataset (*resp.* FLARE 2021). Additionally, it is noteworthy that the proposed method based on U-Net also significantly outperforms SAUNet [44] and TransUNet [12] under cross-dataset evaluation. The proposed method, based on the baseline model SAUNet [44], demonstrates notable enhancements with 20.6% (*resp.* 13.5%) IoU and 19.7% (*resp.* 10.2%) improvement in Dice score on AbdomenCT-1K dataset (*resp.* FLARE 2021). In comparison to SAUNet, which incorporates shape information, the proposed method achieves a further improvement of 16.1% (*resp.* 12.0%) IoU and 15.5% (*resp.* 9.8%) in Dice score on AbdomenCT-1K dataset (*resp.* FLARE 2021). The proposed method based on the latest method LM-Net [50] and 2D D-LKA Net [51] also results in satisfactory generalization effects. The results of our method exhibit a smaller standard deviation, demonstrating greater stability in our approach. Some qualitative results are illustrated in Fig. 4. The proposed SIG-UNet accurately segments the organs.

4.3.2. Polyp segmentation

The quantitative comparison with the baseline models and some state-of-the-art methods is shown in Table 2. The proposed method exhibits improved performance compared to the corresponding baseline on Kvasir and CVC-ClinicDB datasets. Specifically, SIG-UNet built on U-Net [11] improves the baseline U-Net by 1.6% and 3.3% IoU (*resp.* 1.3% and 3.0% Dice score) on Kvasir and CVC-ClinicDB, respectively. Implementing SIG-UNet with the baseline in SAUNet [44], SIG-UNet achieves 2.7% and 0.6% IoU (*resp.* 2.6% and 0.4% Dice score) improvement on Kvasir and CVC-ClinicDB, respectively. Compared with SAUNet which incorporates shape information, SIG-UNet further improved the results. Implementing SIG-UNet with the state-of-the-art method SANet [49] which relies on color exchange, shallow attention module, and probability correction strategy, further boosts the segmentation performance of SANet by 0.6% IoU and 0.3% Dice score on Kvasir dataset, and by 2.1% IoU and 1.2% Dice score on CVC-ClinicDB dataset.

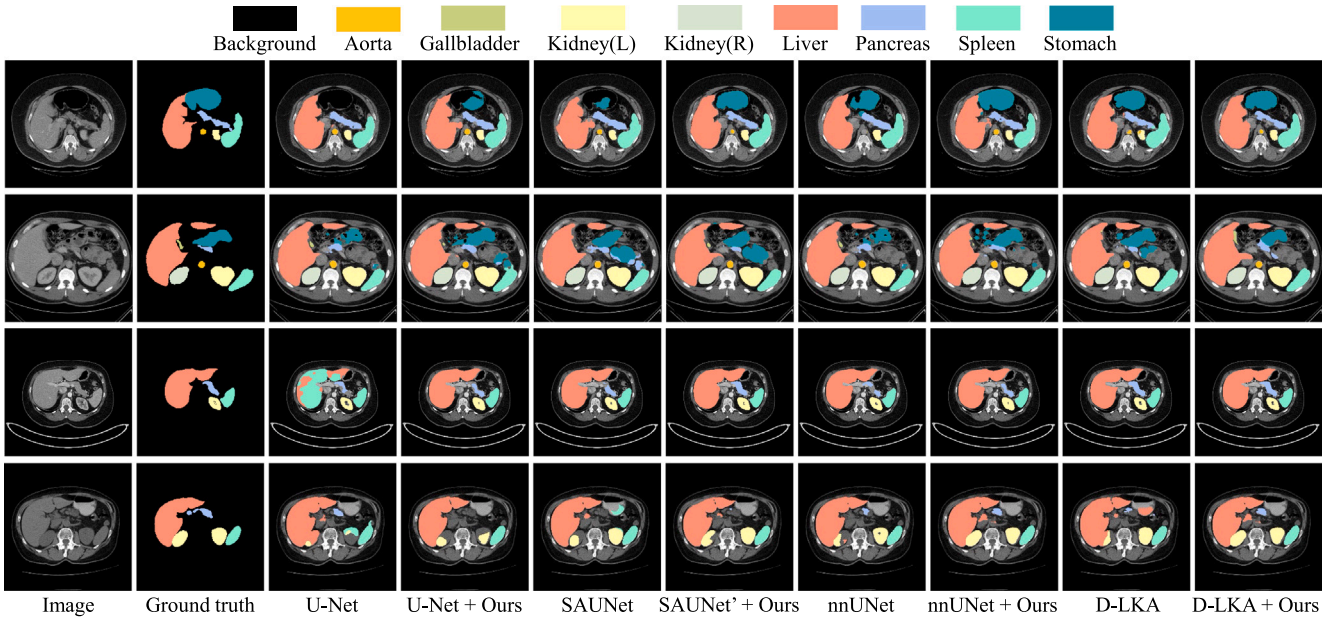


Fig. 4. Qualitative illustration of SIG-UNet and several methods for multi-organ segmentation in CT images on the synapse multi-organ segmentation testing set (Top two rows), the AbdomenCT-1K dataset [70] (The third row) and FLARE 2021 [71] (Bottom row).

Table 2
Quantitative evaluation of polyp segmentation under intra-dataset setting on Kvasir [73], CVC-ClinicDB [74].

Method	Kvasir		CVC-ClinicDB	
	IoU↑	DICE↑	IoU↑	DICE↑
Att-UNet [36]	0.766 ± 0.06	0.848 ± 0.007	0.755 ± 0.007	0.831 ± 0.007
ResUnet [81]	0.788 ± 0.005	0.853 ± 0.006	0.833 ± 0.006	0.884 ± 0.007
PraNet [72]	0.834 ± 0.006	0.892 ± 0.007	0.859 ± 0.005	0.909 ± 0.006
U-Net [11]	0.790 ± 0.005	0.857 ± 0.006	0.821 ± 0.006	0.877 ± 0.008
SIG-UNet (Ours)	0.806 ± 0.003	0.870 ± 0.003	0.854 ± 0.005	0.907 ± 0.006
SAUNet' [44]	0.804 ± 0.005	0.863 ± 0.006	0.812 ± 0.003	0.870 ± 0.004
SAUNet [44]	0.808 ± 0.006	0.866 ± 0.007	0.809 ± 0.005	0.863 ± 0.005
SIG-SAUNet' (Ours)	0.831 ± 0.004	0.889 ± 0.005	0.818 ± 0.002	0.874 ± 0.003
nnUNet [37]	0.822 ± 0.003	0.884 ± 0.005	0.780 ± 0.005	0.837 ± 0.006
SIG-nnUNet (Ours)	0.837 ± 0.003	0.895 ± 0.003	0.822 ± 0.003	0.876 ± 0.004
SANet [49]	0.845 ± 0.005	0.903 ± 0.006	0.861 ± 0.004	0.916 ± 0.005
SIG-SANet (Ours)	**0.851 ± 0.003**	**0.906 ± 0.004**	**0.882 ± 0.004**	**0.928 ± 0.005**
LM-Net [50]	0.827 ± 0.005	0.893 ± 0.007	0.817 ± 0.004	0.872 ± 0.006
SIG-LMNet (Ours)	0.839 ± 0.006	0.901 ± 0.005	0.835 ± 0.003	0.892 ± 0.004
2D D-LKA Net [51]	0.833 ± 0.004	0.890 ± 0.007	0.823 ± 0.004	0.878 ± 0.005
2D SIG-DLKA (Ours)	0.839 ± 0.005	0.899 ± 0.007	0.838 ± 0.003	0.895 ± 0.003

To test the generalizability of the proposed method, we evaluate it on three datasets (CVC-ColonDB, ETIS, and CVC-T) that were not involved during training. As depicted in Table 3, the proposed method based on the U-Net, the baseline of SAUNet [44], nnU-Net, SANet [49], LM-Net [50] and 2D D-LKA Net [51], consistently or significantly boosts the performance on the three unseen datasets, demonstrating good generalization ability. Regardless of the implementation method, our approach consistently demonstrates a smaller standard deviation compared to the corresponding baseline methods. This not only proves that our method offers superior segmentation and generalization performance, but it also exhibits enhanced stability. Fig. 5 showcases some qualitative results of polyp segmentation, where the SIG-UNet achieves accurate segmentation of polyps.

Our approach, when implemented based on different methods, consistently enhances both the segmentation performance and the generalizability of the corresponding methods. This demonstrates our method can be readily integrated into existing methods to improve their efficacy.

4.3.3. *Breast tumor segmentation*

We also adopt the U-Net [11], nnU-Net [37], the baseline of SAUNet [44] and SANet [49] as the baseline to build the proposed method for breast tumor segmentation in ultrasound images. The quantitative benchmark of different methods is depicted in Table 4. Specifically, The proposed approach consistently achieves improvements over the baseline models, with an increase of 1.9% IoU (*resp.* 1.8% Dice score) for U-Net, 6.2% IoU (*resp.* 4.9% Dice score) for the baseline of SAUNet, 0.5% IoU (*resp.* 0.5% Dice score) for nnU-Net, and 1.3% IoU (*resp.* 1.8% Dice score) for SANet.

The generalizability of the proposed method is also assessed by evaluating it on images from the unseen dataset used in [79] and the dataset B [80]. As shown in Table 4, it can be seen that when using U-Net as the baseline model, SIG-UNet achieves 16.2% and 4.1% improvement in terms of IoU (*resp.* 17.4% and 5.3% in terms of Dice score) on the dataset of [79] and the dataset B [80], respectively. The proposed method based on the latest method LM-Net [50] and 2D D-LKA Net [51] also shows consistent improvement under cross-dataset

Table 3
Quantitative evaluation of cross-dataset polyp segmentation on CVC-ColonDB [75], ETIS [76], CVC-T [77] dataset.

Method	CVC-ColonDB		ETIS		CVC-T	
	IoU↑	DICE↑	IoU↑	DICE↑	IoU↑	DICE↑
Att-UNet [36]	0.528 ± 0.011	0.604 ± 0.013	0.476 ± 0.013	0.561 ± 0.014	0.707 ± 0.010	0.792 ± 0.011
ResUnet [81]	0.566 ± 0.010	0.635 ± 0.012	0.512 ± 0.014	0.579 ± 0.016	0.689 ± 0.010	0.773 ± 0.012
PraNet [72]	0.649 ± 0.008	0.721 ± 0.009	0.579 ± 0.012	0.653 ± 0.014	0.818 ± 0.009	0.891 ± 0.009
U-Net [11]	0.538 ± 0.011	0.612 ± 0.012	0.351 ± 0.015	0.407 ± 0.016	0.685 ± 0.011	0.771 ± 0.012
SIG-UNet (Ours)	0.605 ± 0.009	0.688 ± 0.010	0.443 ± 0.012	0.509 ± 0.013	0.703 ± 0.008	0.791 ± 0.008
SAUNet' [44]	0.577 ± 0.012	0.650 ± 0.013	0.450 ± 0.014	0.514 ± 0.015	0.769 ± 0.012	0.843 ± 0.013
SAUNet [44]	0.574 ± 0.011	0.647 ± 0.012	0.493 ± 0.012	0.559 ± 0.013	0.761 ± 0.010	0.830 ± 0.011
SIG-SAUNet' (Ours)	0.581 ± 0.009	0.653 ± 0.011	0.549 ± 0.010	0.617 ± 0.011	0.784 ± 0.008	0.857 ± 0.009
nnUNet [37]	0.645 ± 0.009	0.728 ± 0.010	0.485 ± 0.013	0.573 ± 0.014	0.749 ± 0.009	0.839 ± 0.010
SIG-nnUNet (Ours)	0.668 ± 0.008	0.748 ± 0.009	0.497 ± 0.010	0.587 ± 0.010	0.768 ± 0.007	0.850 ± 0.007
SANet [49]	0.679 ± 0.010	0.762 ± 0.011	0.642 ± 0.011	0.741 ± 0.012	0.785 ± 0.010	0.870 ± 0.011
SIG-SANet (Ours)	**0.685 ± 0.006**	**0.765 ± 0.006**	**0.719 ± 0.009**	**0.801 ± 0.010**	**0.826 ± 0.008**	**0.897 ± 0.009**
LM-Net [50]	0.653 ± 0.010	0.734 ± 0.011	0.565 ± 0.015	0.645 ± 0.014	0.752 ± 0.010	0.835 ± 0.011
SIG-LMNet (Ours)	**0.685 ± 0.007**	0.764 ± 0.009	0.582 ± 0.012	0.663 ± 0.011	0.787 ± 0.008	0.860 ± 0.007
2D D-LKA Net [51]	0.664 ± 0.009	0.742 ± 0.010	0.523 ± 0.014	0.599 ± 0.012	0.771 ± 0.011	0.832 ± 0.011
2D SIG-DLKA (Ours)	0.679 ± 0.011	0.757 ± 0.011	0.553 ± 0.012	0.619 ± 0.013	0.791 ± 0.008	0.862 ± 0.009

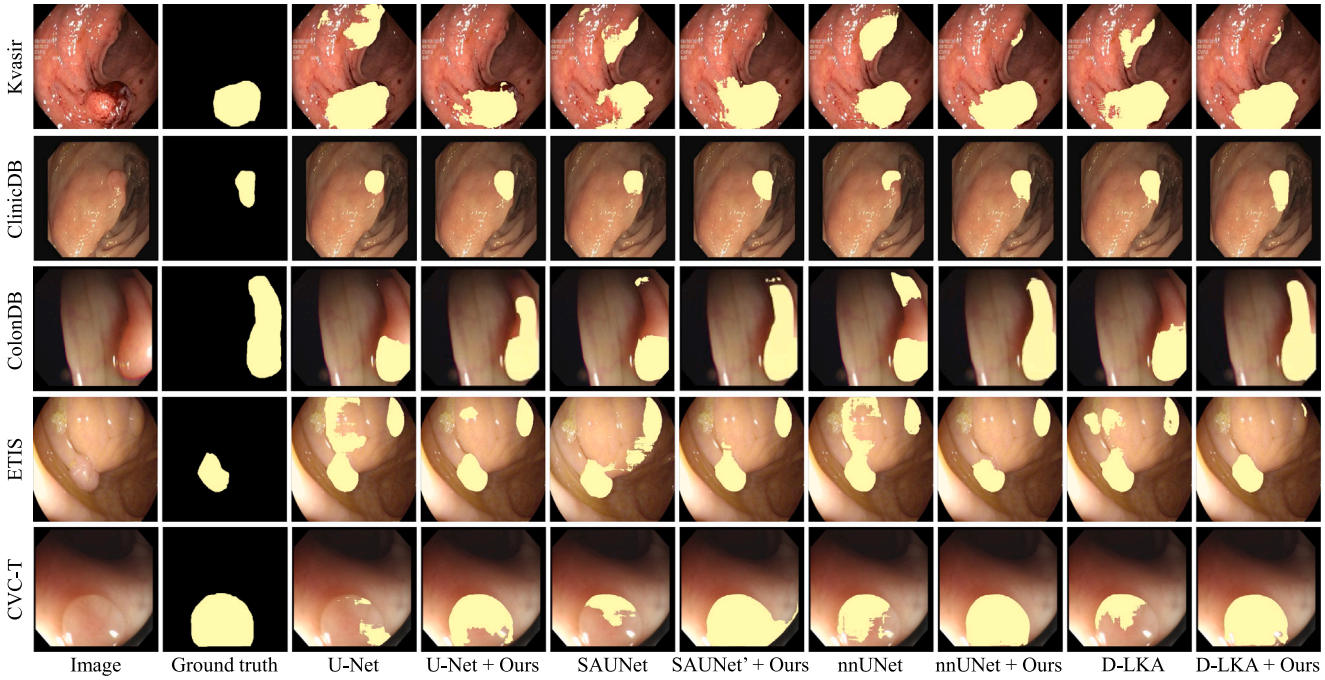


Fig. 5. Visual comparison of segmentation results from different methods, including our approach based on the corresponding method, for intra-domain (first two rows) and cross-domain (last three rows) polyp segmentation.

evaluation. Our method achieved results with lower standard deviation, indicating that it is more stable and thus more suitable for clinical medical applications. Some qualitative results are shown in Fig. 6, showing that the SIG-UNet produces more anatomically consistent, complete, and continuous segmentation results.

4.3.4. Ablation study

We proceed to conduct three types of ablation studies on breast tumor segmentation.

Ablation study on using different reconstruction target: To overcome the network bias towards learning texture information, the proposed method reconstructs a class-wisely averaged image to extract shape and intensity features. A straightforward alternative of mean image reconstruction is to predict the ground-truth segmentation or corresponding edge, which contains only shape information. As depicted in Table 5, extracting and incorporating shape features into segmentation in these two alternatives is also beneficial. The proposed method outperforms these two alternatives in both intra-dataset and cross-dataset evaluations, validating its greater effectiveness. We also conduct an ablation to verify the effectiveness of our dual-decoder structure by utilizing a single decoder network where one head is dedicated to segmentation and the other to reconstructing the mean image. As shown in the results for the "Mean image (Two heads)" in Table 5, it demonstrated an improvement over the baseline model. However, the performance did not reach the level achieved by our proposed dual-decoder structure. This outcome highlights the advantages of our dual-decoder architecture, which facilitates specialized learning and integrates shape-intensity information effectively. This design significantly enhances the network's generalization ability by minimizing texture biases.

Ablation study on using shape-intensity attention block: To verify the effectiveness of using shape-intensity attention block (SIAB) for incorporating shape and intensity features, we follow the original U-Net architecture using direct concatenation in the skip connection between

Table 4
Quantitative evaluation of different methods for breast tumor segmentation on BUSI [78], dataset in [79], and dataset B [80].

Method	Intra-dataset		Cross-dataset			
	BUSI		Dataset in [79]		Dataset B [80]	
	IoU↑	DICE↑	IoU↑	DICE↑	IoU↑	DICE↑
Att-UNet [36]	0.614 ± 0.014	0.704 ± 0.016	0.567 ± 0.016	0.642 ± 0.017	0.505 ± 0.013	0.589 ± 0.013
ResUnet [81]	0.622 ± 0.011	0.716 ± 0.013	0.637 ± 0.013	0.729 ± 0.016	0.659 ± 0.011	0.744 ± 0.013
PraNet [72]	0.649 ± 0.008	0.721 ± 0.009	0.579 ± 0.012	0.653 ± 0.014	0.818 ± 0.009	0.891 ± 0.009
U-Net [11]	0.633 ± 0.013	0.726 ± 0.014	0.589 ± 0.017	0.671 ± 0.018	0.475 ± 0.012	0.556 ± 0.013
SIG-UNet (Ours)	0.652 ± 0.010	0.744 ± 0.011	0.751 ± 0.014	0.845 ± 0.015	0.516 ± 0.009	0.609 ± 0.010
SAUNet' [44]	0.584 ± 0.014	0.687 ± 0.015	0.672 ± 0.015	0.787 ± 0.016	0.551 ± 0.013	0.656 ± 0.014
SAUNet [44]	0.608 ± 0.011	0.708 ± 0.012	0.714 ± 0.013	0.817 ± 0.014	0.535 ± 0.011	0.633 ± 0.012
SIG-SAUNet' (Ours)	0.646 ± 0.008	0.736 ± 0.009	0.722 ± 0.010	0.826 ± 0.012	0.603 ± 0.008	0.691 ± 0.008
nnUNet [37]	0.667 ± 0.013	0.747 ± 0.014	0.433 ± 0.014	0.567 ± 0.015	0.673 ± 0.009	0.761 ± 0.010
SIG-nnUNet (Ours)	0.672 ± 0.011	0.752 ± 0.012	0.612 ± 0.012	0.725 ± 0.013	0.684 ± 0.009	0.770 ± 0.010
SANet [49]	0.676 ± 0.013	0.756 ± 0.014	0.746 ± 0.011	0.841 ± 0.012	0.677 ± 0.012	0.764 ± 0.013
SIG-SANet (Ours)	0.689 ± 0.011	0.774 ± 0.012	0.770 ± 0.008	0.864 ± 0.008	0.679 ± 0.009	0.766 ± 0.010
LM-Net [50]	0.663 ± 0.014	0.755 ± 0.014	0.737 ± 0.013	0.842 ± 0.016	0.571 ± 0.010	0.663 ± 0.014
SIG-LMNet (Ours)	0.690 ± 0.012	0.776 ± 0.009	**0.772 ± 0.013**	**0.865 ± 0.013**	0.631 ± 0.008	0.724 ± 0.013
2D D-LKA Net [51]	0.674 ± 0.013	0.762 ± 0.012	0.593 ± 0.012	0.719 ± 0.015	0.671 ± 0.010	0.760 ± 0.012
2D SIG-DLKA (Ours)	**0.691 ± 0.011**	**0.778 ± 0.010**	0.644 ± 0.013	0.759 ± 0.013	**0.689 ± 0.008**	**0.777 ± 0.011**

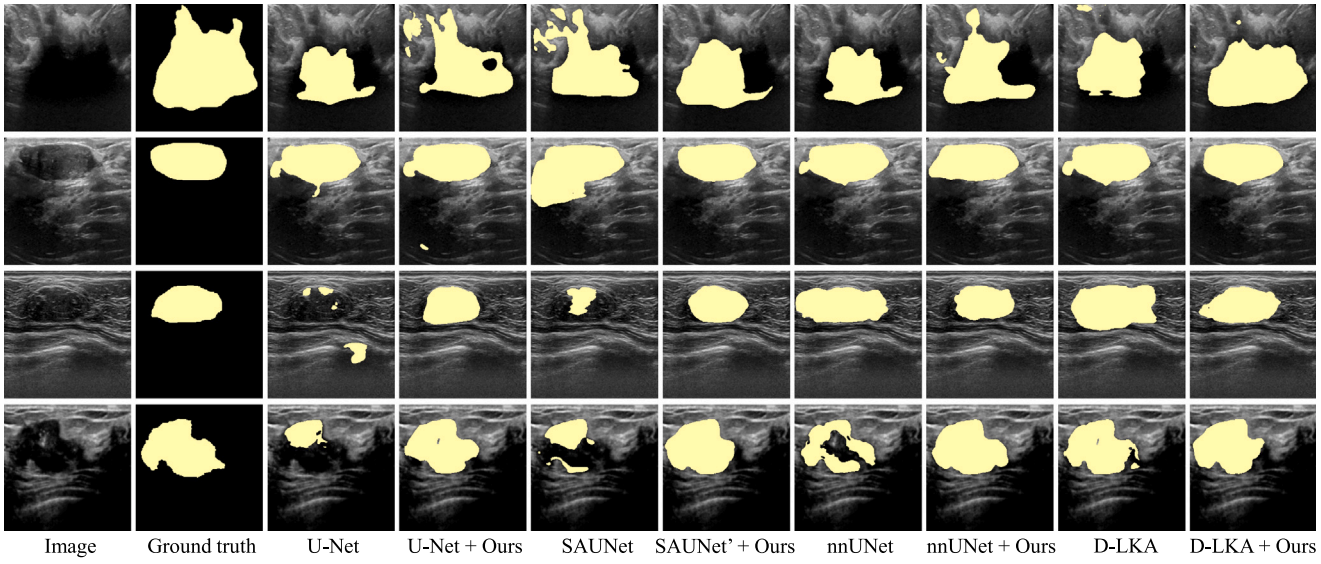


Fig. 6. Visual comparison of segmentation results produced by different methods for breast tumor segmentation on BUSI [78] (Top two row), dataset in [79] (The third row), and dataset B [80] (Bottom row).

Table 5
Quantitative results of the proposed method on BUSI [78], dataset in [79], and dataset B [80] using different learning targets for the reconstruction branch.

Method	Intra-dataset		Cross-dataset			
	BUSI		Dataset in [79]		Dataset B [80]	
	IoU↑	DICE↑	IoU↑	DICE↑	IoU↑	DICE↑
U-Net [11]	0.633 ± 0.013	0.726 ± 0.014	0.589 ± 0.017	0.671 ± 0.018	0.475 ± 0.012	0.556 ± 0.013
Ground truth	0.646 ± 0.013	0.730 ± 0.016	0.639 ± 0.013	0.736 ± 0.011	0.498 ± 0.008	0.588 ± 0.008
Edge	0.647 ± 0.015	0.733 ± 0.018	0.726 ± 0.013	0.831 ± 0.019	0.463 ± 0.010	0.563 ± 0.008
Mean image (Two heads)	0.641 ± 0.011	0.729 ± 0.013	0.628 ± 0.009	0.707 ± 0.014	0.496 ± 0.013	0.587 ± 0.017
SIG-UNet (Ours)	**0.652 ± 0.010**	**0.744 ± 0.011**	**0.751 ± 0.014**	**0.845 ± 0.015**	**0.516 ± 0.009**	**0.609 ± 0.010**

Table 6
Ablation study of SIAB module in the proposed method on BUSI [78], dataset in [79], and dataset B [80].

Method	Intra-dataset		Cross-dataset			
	BUSI		Dataset in [79]		Dataset B [80]	
	IoU↑	DICE↑	IoU↑	DICE↑	IoU↑	DICE↑
U-Net [11]	0.633 ± 0.013	0.726 ± 0.014	0.589 ± 0.017	0.671 ± 0.018	0.475 ± 0.012	0.556 ± 0.013
SIG-UNet wo. SIAB	0.637 ± 0.013	0.732 ± 0.014	0.715 ± 0.015	0.814 ± 0.017	0.497 ± 0.009	0.594 ± 0.012
SIG-UNet w. SIAB (Non-attention)	0.646 ± 0.012	0.738 ± 0.012	0.731 ± 0.015	0.826 ± 0.016	0.501 ± 0.010	0.600 ± 0.013
SIG-UNet w. SIAB	**0.652 ± 0.010**	**0.744 ± 0.011**	**0.751 ± 0.014**	**0.845 ± 0.015**	**0.516 ± 0.009**	**0.609 ± 0.010**

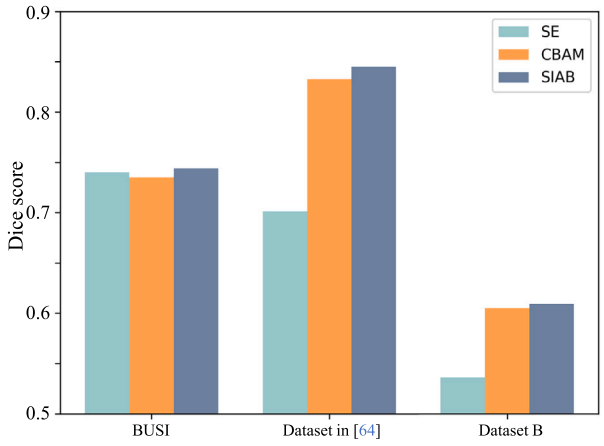


Fig. 7. Comparison of the SIAB module's performance against SE and CBAM.

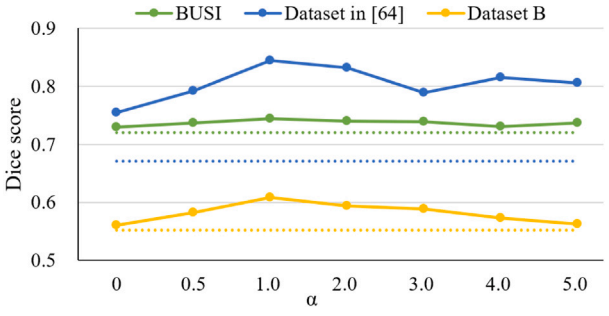


Fig. 8. Evaluation of SIG-UNet built on U-Net with different settings for α in Eq. (5). Setting $\alpha = 0$ indicates the absence of a reconstruction target, and the dashed line represents the corresponding baseline results.

the reconstruction branch and segmentation branch. It can be observed in Table 6 that directly concatenating the features also improves the baseline results, indicating that the fusion of such shape intensity information contributes to better segmentation. It is noteworthy that incorporating such shape and intensity features using the SIAB module further improves the segmentation performance. We further validated the effects of the channel and spatial attention modules within the SIAB module. In Table 6, under "SIG-UNet w. SIAB (Non-attention)", the results indicate that the performance of the model declines when these attention structures are absent, confirming that the channel and spatial attention modules more effectively focus on and integrate shape-intensity information. We also compared the performance of the SIAB module with existing attention modules, SE [83] and CBAM [84]. As illustrated in Fig. 7, the results indicate that our proposed SIAB achieves the best outcomes in both the source and target domains.

Ablation study on the loss weight α: The only hyper-parameter is the loss weight α mentioned in Eq. (5). To analyze the impact of this hyper-parameter, we conducted an ablation study by setting α to different values: 0, 0.5, 1.0, 2.0, 3.0, 4.0, and 5.0 (Setting $\alpha = 0$ indicates the absence of a reconstruction target). The corresponding results are presented in Fig. 8. The intra-dataset performance is less affected by the choice of α. However, the generalization ability exhibits more notable changes. SIG-UNet consistently outperforms the baseline model across various α settings, further proving the effectiveness of the proposed approach. Notably, setting $\alpha = 1$ yields the highest Dice score for breast tumor segmentation. Hence, we select $\alpha = 1$ as the optimal value for the proposed method in all our experiments. Noteworthy, when there is no reconstruction target, the model still performs better than the baseline. This improvement is attributed to the addition of a new branch, which introduces more parameters. This is further elaborated in the discussion section of the manuscript.

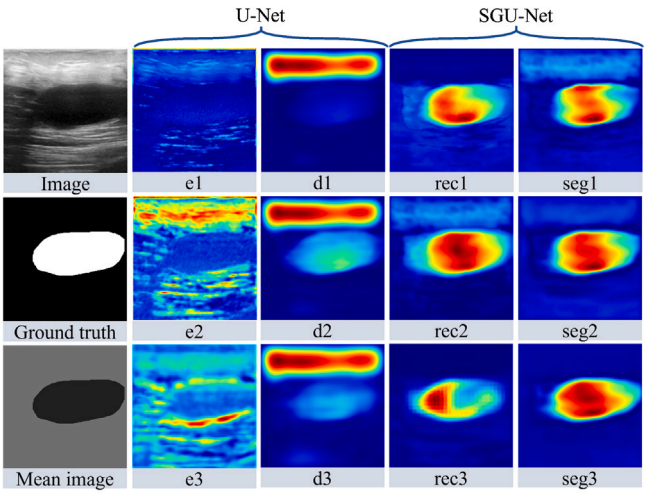


Fig. 9. Visualization of the corresponding layer's features in both the baseline U-Net and the proposed SIG-UNet. e*i*: the *i*th feature of the encoder in U-Net; d*i*: corresponding *i*th feature of the decoder in U-Net; rec*i*: *i*th feature of the reconstruction branch in SIG-UNet; seg*i*: *i*th feature of the segmentation branch in SIG-UNet.

5. Discussion

Convolutional neural networks (CNNs) tend to focus more on learning texture information, as demonstrated in [39,40]. While texture information plays a crucial role in many vision tasks, it presents challenges in the context of medical images due to the wide variation in texture across different datasets. Such texture bias may lead the CNN-based segmentation methods to overfit dataset-specific texture patterns, resulting in poor generalization performance on unseen data. In the field of medical image segmentation, object of interests often have some prior anatomical structures. This motivates a lot of methods in incorporating shape information into segmentation for improving the segmentation performance. Many existing methods (see Section 2.2 for a short review) either introduce shape-related constraints (*e.g.* edge constraint or post-processing using auto-encoder for refinement) or employ adversarial learning. These methods achieve improved segmentation results by such shape information incorporation. Yet, the texture bias issue of CNNs still remains challenging for medical image segmentation.

Different from most existing methods, the proposed method aims to directly incorporate texture inattentive feature for segmentation. Specifically, we adopt the original U-Net to reconstruct class-wisely averaged images, which mainly contain shape and intensity information. Such texture inattentive feature of the reconstruction branch is then fused to an additional segmentation decoder. During the fusion process, instead of a simple skip connection, we leverage the shape-intensity attention block to better incorporate the shape and intensity feature into segmentation. While dual decoder structures are utilized in various networks, they are primarily developed for tasks such as image reconstruction [85] or edge prediction of targets [86]. Our method extends beyond merely adding a reconstruction branch. Importantly, to the best of our knowledge, this is the first proposal to generate class-wisely averaged images containing only shape and intensity information by calculating the mean pixel value of the target. The proposed method directly and effectively alleviates the texture bias issue, giving rise to improved segmentation accuracy and generalization ability.

We conduct a specific experiment on BUSI dataset [78] to evaluate the effectiveness of the proposed method in alleviating texture bias issue. More precisely, we synthetically enhance the texture information by weighted sum of the original image and corresponding result of Sobel operator (extracting high frequency information). Then, we

Table 7
Quantitative results of the proposed method on texture enhanced BUSI [78] images given by weighted sum of original image and corresponding result f_s of Sobel operator.

Method	Orinal Image f		$0.5*f + 0.5*f_s$	
	IoU↑	DICE↑	IoU	DICE
U-Net [11]	0.633 ± 0.013	0.726 ± 0.014	0.406 ± 0.019	0.514 ± 0.021
SIG-UNet (Ours)	**0.652 ± 0.010**	**0.744 ± 0.011**	**0.515 ± 0.014**	**0.620 ± 0.015**

compare the results of SIG-UNet and the baseline U-Net (trained on the original training dataset) on the texture enhanced test images. As shown in Table 7, the Dice score of SIG-UNet (*resp.* U-Net) drops from 0.744 to 0.620 (*resp.* 0.726 to 0.514). The gap between SIG-UNet and U-Net increases from 1.8% to 10.8%, showing that SIG-UNet is more robust to texture changes. This experiment further demonstrates that the proposed method is effective in mitigating texture bias issue.

We also visualize the learned feature to better analyze the effect of the proposed method in alleviating texture bias. As illustrated in Fig. 9, the shallow encoder features (second column) of U-Net focus predominantly on the texture information in the input image. Consequently, the deep decoder features (the third column) struggle to accurately focus on the correct target location. In contrast, for the proposed method, the features (the fourth column) of the reconstruction branch is less bias towards texture. Incorporating such shape feature into segmentation makes the network focus more on the target region, yielding improved segmentation performance and generalizability.

In short, this paper aims to directly address the texture bias issue of CNNs in medical image segmentation. By reconstructing class-wisely averaged images to extract shape and intensity feature and incorporating such feature into segmentation, the proposed method contributes to the development of more robust and accurate medical image segmentation techniques for better and more reliable clinical decision-making. Our method enhances model generalization by reducing texture bias, promising improved performance across multiple imaging modalities. Although the experiments presented in our manuscript were validated on CT, colonoscopy, and ultrasound images, our method is equally applicable to other modalities, such as MRI. This adaptability is crucial for clinical practice, where variations in imaging conditions and modalities are prevalent. A limitation of the proposed method is the additional computational time required during the training phase due to the need to calculate the class-wisely averaged image. Furthermore, the incorporation of the reconstruction branch and the shape-intensity attention module inevitably increases the number of parameters in our model.

6. Conclusion

This paper presents the Shape-Intensity-Guided U-Net (SIG-UNet), a novel approach that addresses the issue of network bias towards texture features, resulting in improved segmentation results and remarkable generalization ability. This is achieved by utilizing a two-branch decoder structure. Firstly, we adopt the original U-Net architecture to reconstruct a class-wisely averaged image containing no texture information. This enables the network to focus mainly on extracting shape feature extraction. Secondly, we add an extra branch for the segmentation task, consisting of the shape-intensity attention block that effectively fuses the shape feature from the reconstruction branch into segmentation. Extensive experiments on various medical image segmentation tasks with different modalities demonstrate that the proposed method consistently or significantly outperforms the corresponding baseline methods.

In future research, we aim to integrate adaptive learning techniques with our proposed approach to further enhance the model's generalization capabilities. This combination is designed to adaptively adjust the model parameters to better accommodate variations in data collected from different centers or devices. Additionally, large model technologies have seen rapid advancements in the past two years, such as Large Language Models (LLMs) and Vision-Language Models (VLMs). In future work, we intend to explore the incorporation of prior information, such as language descriptions of target shapes and intensities, into these large models to further enhance their generalization capabilities.

CRediT authorship contribution statement

Wenhui Dong: Writing – review & editing. **Bo Du:** Supervision. **Yongchao Xu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Key Research and Development Program of China (2023YFC2705700), NSFC 62222112, and 62176186, the Innovative Research Group Project of Hubei Province under Grants (2024AFA017), and CAAI Huawei MindSpore Open Fund.

References

[1] Y. Zhou, X. Kang, F. Ren, H. Lu, S. Nakagawa, X. Shan, A multi-attention and depthwise separable convolution network for medical image segmentation, Neurocomputing 564 (2024) 126970.
[2] S. Lu, Z. Zhang, Z. Yan, Y. Wang, T. Cheng, R. Zhou, G. Yang, Mutually aided uncertainty incorporated dual consistency regularization with pseudo label for semi-supervised medical image segmentation, Neurocomputing 548 (2023) 126411.
[3] B. Rahmati, S. Shirani, Z. Keshavarz-Motamed, Redundant co-training: Semi-supervised segmentation of medical images using informative redundancy, Neurocomputing 579 (2024) 127446.
[4] X. Cui, S. Chang, C. Li, B. Kong, L. Tian, H. Wang, P. Huang, M. Yang, Y. Wu, Z. Li, DEAttack: A differential evolution based attack method for the robustness evaluation of medical image segmentation, Neurocomputing 465 (2021) 38–52.
[5] H. Wang, Z. Wang, X. Wang, Z. Wu, Y. Yuan, Q. Li, AnatSwin: An anatomical structure-aware transformer network for cardiac MRI segmentation utilizing label images, Neurocomputing 577 (2024) 127379.
[6] M. Lin, Q. Cai, J. Zhou, 3D Md-Unet: A novel model of multi-dataset collaboration for medical image segmentation, Neurocomputing 492 (2022) 530–544.
[7] Y. Ge, Z.-M. Chen, G. Zhang, A.A. Heidari, H. Chen, S. Teng, Unsupervised domain adaptation via style adaptation and boundary enhancement for medical semantic segmentation, Neurocomputing 550 (2023) 126469.
[8] B. Wang, Q. Li, Z. You, Self-supervised learning based transformer and convolution hybrid network for one-shot organ segmentation, Neurocomputing 527 (2023) 1–12.
[9] X. Hu, Z. Li, Y. Wu, J. Liu, X. Luo, J. Ren, Neighbouring-slice guided multi-view framework for brain image segmentation, Neurocomputing 575 (2024) 127315.
[10] J. Wu, D. Guo, L. Wang, S. Yang, Y. Zheng, J. Shapey, T. Vercauteren, S. Bisdas, R. Bradford, S. Saeed, N. Kitchen, S. Ourselin, S. Zhang, G. Wang, TISS-net: Brain tumor image synthesis and segmentation using cascaded dual-task networks and error-prediction consistency, Neurocomputing 544 (2023) 126295.
[11] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2015, pp. 234–241.

[12] J. Chen, et al., TransUNet: Transformers make strong encoders for medical image segmentation, 2021, arXiv.

[13] H. Cao, et al., Swin-unet: Unet-like pure transformer for medical image segmentation, in: ECCV Workshops, 2022, pp. 205–218.

[14] F. Milletari, N. Navab, S.-A. Ahmadi, V-Net: Fully convolutional neural networks for volumetric medical image segmentation, in: 2016 Fourth International Conference on 3D Vision, 3DV, 2016, pp. 565–571.

[15] Ö. Çiçek, A. Abdulkadir, S.S. Lienkamp, T. Brox, O. Ronneberger, 3D U-Net: learning dense volumetric segmentation from sparse annotation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2016, pp. 424–432.

[16] Q. Jin, Z. Meng, C. Sun, H. Cui, R. Su, RA-UNet: A hybrid deep attention-aware network to extract liver and tumor in CT scans, Front. Bioeng. Biotechnol. 8 (2020) 605132.

[17] Z. Zhou, M.M.R. Siddiquee, N. Tajbakhsh, J. Liang, Unet++: Redesigning skip connections to exploit multiscale features in image segmentation, IEEE Trans. Med. Imag. 39 (6) (2019) 1856–1867.

[18] Q. Huang, H. Tian, L. Jia, Z. Li, Z. Zhou, A review of deep learning segmentation methods for carotid artery ultrasound images, Neurocomputing 545 (2023) 126298.

[19] Q. Huang, L. Zhao, G. Ren, X. Wang, C. Liu, W. Wang, NAG-Net: Nested attention-guided learning for segmentation of carotid lumen-intima interface and media-adventitia interface, Comput. Biol. Med. 156 (2023) 106718.

[20] Q. Huang, L. Jia, G. Ren, X. Wang, C. Liu, Extraction of vascular wall in carotid ultrasound via a novel boundary-delineation network, Eng. Appl. Artif. Intell. 121 (2023) 106069.

[21] G. Li, Q. Huang, C. Liu, G. Wang, L. Guo, R. Liu, L. Liu, Fully automated diagnosis of thyroid nodule ultrasound using brain-inspired inference, Neurocomputing 582 (2024) 127497.

[22] E.M. Elmeslimany, S.S. Kishk, D.A. Altantawy, ΨNet: a parallel network with deeply coupled spatial and squeezed features for segmentation of medical images, Multimedia Tools Appl. 83 (8) (2024) 24045–24082.

[23] M.S. Sheela, G. Amirthayogam, J.J. Hephzipah, S. Gopalakrishnan, S.R. Chand, Machine learning based lung disease prediction using convolutional neural network algorithm, Mesop. J. Artif. Intell. Healthc. 2024 (2024) 50–58.

[24] R. Rajkumar, S. Gopalakrishnan, K. Praveena, M. Venkatesan, K. Ramamoorthy, J.J. Hephzipah, DARKNET-53 convolutional neural network-based image processing for breast cancer detection, Mesop. J. Artif. Intell. Healthc. 2024 (2024) 59–68.

[25] H. Xu, Z. Chen, Y. Zhang, X. Geng, S. Mi, Z. Yang, Weakly supervised temporal action localization with proxy metric modeling, Front. Comput. Sci. 17 (2) (2023) 172309.

[26] Y. Zhang, Z. Wang, J. Zhou, S. Mi, Person video alignment with human pose registration, Front. Comput. Sci. 17 (4) (2023) 174324.

[27] S. Tan, L. Zhang, X. Shu, Z. Wang, A feature-wise attention module based on the difference with surrounding features for convolutional neural networks, Front. Comput. Sci. 17 (6) (2023) 176338.

[28] M. Guo, H. Sheng, Z. Zhang, Y. Huang, X. Chen, C. Wang, J. Zhang, CW-YOLO: joint learning for mask wearing detection in low-light conditions, Front. Comput. Sci. 17 (6) (2023) 176710.

[29] K. Wu, F. Jia, Y. Han, Domain-specific feature elimination: multi-source domain adaptation for image classification, Front. Comput. Sci. 17 (4) (2023) 174705.

[30] Y. Zhu, X. Wu, J. Qiang, Y. Yuan, Y. Li, Representation learning via an integrated autoencoder for unsupervised domain adaptation, Front. Comput. Sci. 17 (5) (2023) 175334.

[31] Q. Tian, H. Sun, S. Peng, T. Ma, Self-adaptive label filtering learning for unsupervised domain adaptation, Front. Comput. Sci. 17 (1) (2023) 171308.

[32] Z. Ji, J. Ni, X. Liu, Y. Pang, Teachers cooperation: team-knowledge distillation for multiple cross-domain few-shot learning, Front. Comput. Sci. 17 (2) (2023) 172312.

[33] Z. Wu, Y. Gan, T. Xu, F. Wang, Graph-Segmenter: Graph transformer with boundary-aware attention for semantic segmentation, Front. Comput. Sci. 18 (5) (2024) 185327.

[34] H. Ruan, H. Song, B. Liu, Y. Cheng, Q. Liu, Intellectual property protection for deep semantic segmentation models, Front. Comput. Sci. 17 (1) (2023) 171306.

[35] E. Chen, S. Chen, T. Ye, Y. Liu, Degradation-adaptive neural network for jointly single image dehazing and desnowing, Front. Comput. Sci. 18 (2) (2024) 182707.

[36] O. Oktay, et al., Attention U-net: Learning where to look for the pancreas, in: Medical Imaging with Deep Learning, 2018.

[37] F. Isensee, P.F. Jaeger, S.A. Kohl, J. Petersen, K.H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, Nature Methods 18 (2) (2021) 203–211.

[38] D. Gut, Z. Tabor, M. Szymkowski, M. Rozynek, I. Kucybała, W. Wojciechowski, Benchmarking of deep architectures for segmentation of medical images, IEEE Trans. Med. Imag. (2022) 1.

[39] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F.A. Wichmann, W. Brendel, ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness, in: Proc. of International Conference on Learning Representations, 2019.

[40] Y. Li, et al., Shape-texture debiased neural network training, in: Proc. of International Conference on Learning Representations, 2021.

[41] S. Bhojanapalli, A. Chakrabarti, D. Glasner, D. Li, T. Unterthiner, A. Veit, Understanding robustness of transformers for image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10231–10241.

[42] M.M. Naseer, K. Ranasinghe, S.H. Khan, M. Hayat, F. Shahbaz Khan, M.-H. Yang, Intriguing properties of vision transformers, Adv. Neural Inf. Process. Syst. 34 (2021) 23296–23308.

[43] X. Xie, J. Niu, X. Liu, Z. Chen, S. Tang, S. Yu, A survey on incorporating domain knowledge into deep learning for medical image analysis, Med. Image Anal. 69 (2021) 101985.

[44] J. Sun, F. Darbehani, M. Zaidi, B. Wang, SAUnet: Shape attentive u-net for interpretable medical image segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2020, pp. 797–806.

[45] L. Li, V.A. Zimmer, J.A. Schnabel, X. Zhuang, AtrialJSQnet: A new framework for joint segmentation and quantification of left atrium and scars incorporating spatial and shape information, Med. Image Anal. 76 (2022) 102303.

[46] Z. Ning, S. Zhong, Q. Feng, W. Chen, Y. Zhang, SMU-Net: Saliency-guided morphology-aware U-Net for breast lesion segmentation in ultrasound image, IEEE Trans. Med. Imag. 41 (2) (2022) 476–490.

[47] K.B. Girum, G. Créhange, A. Lalande, Learning with context feedback loop for robust medical image segmentation, IEEE Trans. Med. Imag. 40 (6) (2021) 1542–1554.

[48] T. Lei, R. Sun, X. Du, H. Fu, C. Zhang, A.K. Nandi, SGU-Net: Shape-guided ultralight network for abdominal image segmentation, IEEE J. Biomed. Health Inf. 27 (3) (2023) 1431–1442.

[49] J. Wei, Y. Hu, R. Zhang, Z. Li, S.K. Zhou, S. Cui, Shallow attention network for polyp segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2021, pp. 699–708.

[50] Z. Lu, C. She, W. Wang, Q. Huang, LM-Net: A light-weight and multi-scale network for medical image segmentation, Comput. Biol. Med. 168 (2024) 107717.

[51] R. Azad, L. Niggemeier, M. Hüttemann, A. Kazerouni, E.K. Aghdam, Y. Velichko, U. Bagci, D. Merhof, Beyond self-attention: Deformable large kernel attention for medical image segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 1287–1297.

[52] T. Xiang, C. Zhang, D. Liu, Y. Song, H. Huang, W. Cai, BiO-Net: learning recurrent bi-directional connections for encoder-decoder architecture, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2020, pp. 74–84.

[53] M.Z. Alom, M. Hasan, C. Yakopcic, T.M. Taha, V.K. Asari, Recurrent residual convolutional neural network based on u-net (r2u-net) for medical image segmentation, 2018, arXiv.

[54] M. Alom, M. Hasan, C. Yakopcic, T.M. Taha, V.K. Asari, Improved inception-residual convolutional neural network for object recognition, Neural Comput. Appl. 32 (1) (2020) 279–293.

[55] X. Xiao, S. Lian, Z. Luo, S. Li, Weighted res-unet for high-quality retina vessel segmentation, in: Proc. of Intl. Conf. on Information Technology in Medicine and Education, 2018, pp. 327–331.

[56] Z. Yan, X. Yang, K.-T. Cheng, Enabling a single deep learning model for accurate gland instance segmentation: A shape-aware adversarial learning framework, IEEE Trans. Med. Imag. 39 (6) (2020) 2176–2189.

[57] O. Oktay, et al., Anatomically constrained neural networks (ACNNs): application to cardiac image enhancement and segmentation, IEEE Trans. Med. Imag. 37 (2) (2017) 384–395.

[58] N. Painchaud, Y. Skandarani, T. Judge, O. Bernard, A. Lalande, P.-M. Jodoin, Cardiac segmentation with strong anatomical guarantees, IEEE Trans. Med. Imag. 39 (11) (2020) 3703–3713.

[59] A.J. Larrazabal, C. Martínez, B. Glocker, E. Ferrante, Post-DAE: Anatomically plausible segmentation via post-processing with denoising autoencoders, IEEE Trans. Med. Imag. 39 (12) (2020) 3813–3820.

[60] C. Chen, K. Hammernik, C. Ouyang, C. Qin, W. Bai, D. Rueckert, Cooperative training and latent space data augmentation for robust medical image segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2021, pp. 149–159.

[61] X. Yi, E. Walia, P. Babyn, Generative adversarial network in medical imaging: A review, Med. Image Anal. 58 (2019) 101552.

[62] M. Jafari, S. Francis, J.M. Garibaldi, X. Chen, LMISA: A lightweight multi-modality image segmentation network via domain adaptation using gradient magnitude and shape constraint, Med. Image Anal. 81 (2022) 102536.

[63] C. Zotti, Z. Luo, A. Lalande, P.-M. Jodoin, Convolutional neural network with shape prior applied to cardiac MRI segmentation, IEEE J. Biomed. Health Inf. 23 (3) (2018) 1119–1128.

[64] S. Tilborghs, J. Bogaert, F. Maes, Shape constrained CNN for segmentation guided prediction of myocardial shape and pose parameters in cardiac MRI, Med. Image Anal. (2022) 102533.

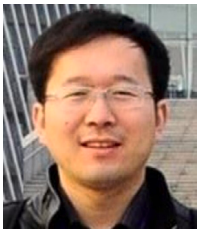
[65] Z. Mirikharaji, G. Hamarneh, Star shape prior in fully convolutional networks for skin lesion segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2018, pp. 737–745.

[66] F. Guo, M. Ng, G. Kuling, G. Wright, Cardiac MRI segmentation with sparse annotations: Ensembling deep learning uncertainty and shape priors, Med. Image Anal. 81 (2022) 102532.

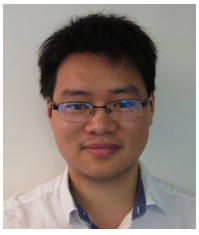
[67] H. Wei, J. Ma, Y. Zhou, W. Xue, D. Ni, Co-learning of appearance and shape for precise ejection fraction estimation from echocardiographic sequences, Med. Image Anal. 84 (2023) 102686.
[68] Z. Zhang, L. Yang, Y. Zheng, Translating and segmenting multimodal medical volumes with cycle-and shape-consistency generative adversarial network, in: Proc. of IEEE Intl. Conf. on Computer Vision and Pattern Recognition, 2018, pp. 9242–9251.
[69] B. Landman, Z. Xu, J. Igelsias, M. Styner, T. Langerak, A. Klein, Miccai multi-atlas labeling beyond the cranial vault–workshop and challenge, in: Proc. of MICCAI Workshop, Vol. 5, 2015, p. 12.
[70] J. Ma, et al., AbdomenCT-1K: Is abdominal organ segmentation a solved problem, IEEE Trans. Pattern Anal. Mach. Intell. 44 (10) (2022) 6695–6714.
[71] J. Ma, Y. Zhang, S. Gu, X. An, Z. Wang, C. Ge, C. Wang, F. Zhang, Y. Wang, Y. Xu, S. Gou, F. Thaler, C. Payer, D. Štern, E.G. Henderson, D.M. McSweeney, A. Green, P. Jackson, L. McIntosh, Q.-C. Nguyen, A. Qayyum, P.-H. Conze, Z. Huang, Z. Zhou, D.-P. Fan, H. Xiong, G. Dong, Q. Zhu, J. He, X. Yang, Fast and low-GPU-memory abdomen CT organ segmentation: The FLARE challenge, Med. Image Anal. 82 (2022) 102616.
[72] D.-P. Fan, et al., PraNet: Parallel reverse attention network for polyp segmentation, in: Proc. of Intl. Conf. on Medical Image Computing and Computer Assisted Intervention, 2020, pp. 263–273.
[73] D. Jha, et al., Kvasir-seg: A segmented polyp dataset, in: International Conference on Multimedia Modeling, 2020, pp. 451–462.
[74] J. Bernal, F.J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez, F. Vilariño, WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians, Comput. Med. Imaging Graph. 43 (2015) 99–111.
[75] N. Tajbakhsh, S.R. Gurudu, J. Liang, Automated polyp detection in colonoscopy videos using shape and context information, IEEE Trans. Med. Imag. 35 (2) (2015) 630–644.
[76] J. Silva, A. Histace, O. Romain, X. Dray, B. Granado, Toward embedded detection of polyps in wce images for early diagnosis of colorectal cancer, Int. J. Comput. Assist. Radiol. Surg. 9 (2) (2014) 283–293.
[77] D. Vázquez, J. Bernal, F.J. Sánchez, G. Fernández-Esparrach, A.M. López, A. Romero, M. Drozdzal, A. Courville, A benchmark for endoluminal scene segmentation of colonoscopy images, J. Healthc. Eng. 2017 (2017).
[78] W. Al-Dhabyani, M. Gomaa, H. Khaled, A. Fahmy, Dataset of breast ultrasound images, Data Brief 28 (2020) 104863.
[79] Z. Zhuang, N. Li, A.N. Joseph Raj, V.G. Mahesh, S. Qiu, An RDAU-NET model for lesion segmentation in breast ultrasound images, PLoS One 14 (8) (2019) e0221535.
[80] M.H. Yap, et al., Automated breast ultrasound lesions detection using convolutional neural networks, IEEE J. Biomed. Health Inf. 22 (2018) 1218–1226.
[81] Z. Zhang, Q. Liu, Y. Wang, Road extraction by deep residual u-net, IEEE Geosci. Remote Sens. Lett. 15 (5) (2018) 749–753.
[82] D.P. Kingma, J. Ba, Adam: a method for stochastic optimization, in: Proc. of International Conference on Learning Representations, 2014, pp. 1–11.
[83] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proc. of IEEE Intl. Conf. on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.
[84] S. Woo, J. Park, J.-Y. Lee, I.S. Kweon, Cbam: Convolutional block attention module, in: Proc. of European Conference on Computer Vision, 2018, pp. 3–19.
[85] C. Chen, Z. Li, C. Ouyang, M. Sinclair, W. Bai, D. Rueckert, Maxstyle: Adversarial style composition for robust medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2022, pp. 151–161.
[86] F. Min, L. Wang, S. Pan, G. Song, D2UNet: Dual decoder U-net for seismic image super-resolution reconstruction, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–13.



Wenhui Dong received the M.S. degree from the School of Software Engineering, Wuhan University, Wuhan, China, in 2020. He is currently pursuing the Ph.D. degree in the School of Computer Science, Wuhan University, Wuhan, China. His main research interests include image segmentation, medical image analysis, and video object segmentation.



Bo Du received the Ph.D. degree in photogrammetry and remote sensing from the State Key Lab of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2010. He is currently a professor and the dean of School of Computer Science. He is also the director of the National Engineering Research Center for Multimedia Software, Wuhan University. His major research interests include machine learning, computer vision, and image processing. He has more than 80 journal papers published in IEEE TPAMI/TIP/TCYB/TGRS, IJCV. He serves as associate editor of Neural Networks, Pattern Recognition, and Neurocomputing. He won the Highly Cited researcher (2019/2020/2021/2022) by the Web of Science Group. He also won IEEE Geoscience and Remote Sensing Society 2020 Transactions Prize Paper Award, and IJCAI Distinguished Paper Prize. He regularly serves as senior PC member of IJCAI and AAAI.



Yongchao Xu received in 2010 the master degree in electronics and signal processing at Université Paris Sud, and the Ph.D. degree in image processing at Université Paris Est in 2013. He is currently a Professor with the School of Computer Science, Wuhan University. His research interests include image segmentation, medical image analysis, cross-domain generalization for deep learning. He has published more than 40 scientific papers, such as IEEE TPAMI, IJCV, IEEE TIP, CVPR, ICCV, ECCV, MICCAI. He serves as associate editor of Image and Vision Computing, and young associate editor of Frontiers of Computer Science.